PathFinder: Project Action Plan & Literature References

1. Week-Wise Project Action Plan

Week 1

Completed: Finalized the project title, defined objectives, studied the existing system, identified limitations, discussed project modules, selected core technologies (Python, Streamlit/Flask, NLP, ChromaDB, SQLite), and collected initial datasets and research papers.

Planned Next Steps: Perform a detailed literature survey, finalize system architecture, prepare software requirements, and create the project workflow.

Week 2

Completed: Conducted literature survey, analyzed IEEE/Springer research papers, finalized methodology, prepared system flow diagrams and database schema design, and created the project repository folder structure.

Planned Next Steps: Set up the development environment, configure environment variables, and begin implementing the frontend interface and local SQLite database.

Week 3

Completed: Developed the initial Streamlit frontend interface (Home, Profile Input, Dashboard), configured SQLite database schema, and installed required Python libraries (pandas, scikit-learn, spacy).

Planned Next Steps: Implement user profile storage, session state management, and input validation.

Week 4

Completed: Implemented user profile submission handling, session persistence, and data validation rules. Configured environment keys and fallback handlers.

Planned Next Steps: Begin implementing resume file upload (PDF/DOCX), stream text extraction, and NLP-based skill extraction.

Week 5

Completed: Implemented resume file upload functionality (PDF/DOCX) using PyPDF and python-docx, and developed NLP-based resume parsing to extract candidate skills, education, and contact details.

Planned Next Steps: Develop the skill extraction normalization module and integrate hybrid career recommendation algorithms.

Week 6

Completed: Implemented skill extraction, hybrid TF-IDF Cosine Similarity career prediction, and LLM-powered recommendation engine using SpaCy and Groq API (Llama-3).

Planned Next Steps: Integrate semantic course recommendations (ChromaDB vector search), certification suggestions, and skill-gap analysis.

Week 7

Completed: Developed certification recommendations, educational course vector retrieval, personalized week-by-week learning roadmaps, and career readiness analysis.

Planned Next Steps: Improve recommendation accuracy using hybrid overlap-cosine scoring and implement the 2030 Future-Proofing Index.

Week 8

Completed: Integrated the 2030 Future-Proofing Index (Career Time Machine) and Direct Job Board matching modules. Enhanced UI aesthetics with modern CSS badges, glassmorphism card layouts, and progress bars.

Planned Next Steps: Perform comprehensive module testing, edge-case error handling validation, and optimize backend performance.

Week 9

Completed: Completed end-to-end system testing, validated resume extraction accuracy across multiple PDF/DOCX layouts, verified API fallback mechanisms, fixed edge-case bugs, and optimized local database queries.

Planned Next Steps: Prepare complete project documentation, user manual, thesis report, presentation slides, and demonstration video.

Week 10

Completed: Completed final documentation, prepared presentation slides, conducted deployment testing on Streamlit Community Cloud, demonstrated the application, and deployed the project for evaluation.

Planned Next Steps: Final project review, viva preparation, and project submission.

2. Literature Survey & Research References (2020–2026)

The following high-quality research papers from reputed publishers (IEEE Transactions, IEEE Access, Elsevier, Springer, arXiv) form the academic foundation for PathFinder's architecture:

Year	Paper Title	Publisher / Journal
2026	AI-Powered Career Guidance: A Scalable Model for Personalized Recommendations	IEEE Access
2026	A Knowledge Graph-Integrated Recommendation Method for College Student Career Planning	Springer (Discover Artificial Intelligence)
2025	CareerBERT: Matching Resumes to ESCO Jobs in a Shared Embedding Space for Generic Job Recommendations	arXiv (Cited in NLP/Job matching)
2023	Efficient Resume-Based Re-Education for Career Recommendation in Rapidly Evolving Job Markets	IEEE Access
2023	Enhancing Job Recommendation through LLM-based Generative Adversarial Networks	arXiv
2020	PCRS: Personalized Career-Path Recommender System for Engineering Students	IEEE Access
2020	Using Autoencoders for Session-Based Job Recommendations	Springer (User Modeling & Interaction)
2020	Competence-Level Prediction and Resume & Job Description Matching Using Context-Aware Transformer Models	arXiv

3. Research Gap & System Contributions

Identified Limitations of Existing Systems:

- Most existing platforms focus solely on basic career prediction or job recommendation without offering an end-to-end learning pathway.
- Traditional solutions perform simple keyword parsing on resumes but fail to provide structured, week-by-week upskilling roadmaps.

- Current platforms do not integrate candidate profile entry, vector search course retrieval, and automation risk analysis in a unified system.
- Existing systems lack dynamic 2030 future-proofing analysis to evaluate skill vulnerability against AI automation.

How PathFinder Addresses the Gap:

- **Dual Input Support:** Seamlessly accepts both manual profile entry and automated PDF/DOCX resume text extraction.
- **NLP & Hybrid Matching:** Combines SpaCy NER, TF-IDF Cosine Similarity, and Llama-3 LLM for objective career alignment.
- **ChromaDB Semantic Search:** Vectorized retrieval of curated courses and professional certifications matching career goals.
- **Personalized Roadmaps:** Generates structured week-by-week learning timelines customized to candidate skill gaps.
- **2030 Future-Proofing Index:** Evaluates skill automation risk and suggests protective 'Shield Skills'.
- **Direct Job Alignment:** Displays real-time job openings with color-coded matched vs. missing skill badges.
- **Failsafe Execution:** Runs on high-speed cloud AI with an automatic, seamless local math fallback on API limits.